Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7.	(a)			$2C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$ correct formulae (1) balancing and state symbols (1)		2		2		
	(b)			any of following for (1) • if carbon, hydrogen and oxygen are reacted other products (as well as ethanol) would form • the activation energy for the reaction is too high • carbon, hydrogen and oxygen do not easily react with each other under these conditions	1			1		
	(c)	(i)		use of energy / $Q = mc\Delta T$ (1) use of moles (ethanol) = $\dfrac{\text{mass of ethanol}}{46/\ M_r}$ (1) $\Delta_c H$ = energy/ moles ethanol (1)		3		3		3
		(ii)		ethane is a gas / practical problem based on ethane being a gas			1	1		1
	(d)			Euan: temperature rise / loss in mass very small therefore greater percentage error / less accurate (1) Carys: mass of fuel used after boiling did not result in temperature rise therefore calculated value of $\Delta_c H$ too small (1)			2	2		2

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
	(e)	(i)		Hess' cycle drawn (1) $\Delta_f H^\ominus - 1371 = 2(-394) + 3(-286)$ (1) $\Delta_f H^\ominus = -275$ (1) award (3) for cao ecf possible		3		3	2	
		(ii)		Amir is correct – no credit award credit only for calculations C 394/12 H_2 286/2.02 C_2H_5OH 1371/46 (1) C 32.8 H_2 142 C_2H_5OH 29.8 (1)			2	2	1	
				Question 7 total	**1**	**8**	**5**	**14**	**3**	**6**

(Marks available: AO1, AO2, AO3, Total, Maths, Prac)